

Efficient Algorithms for Computing the Reliability of Consecutive-k-out-of-n: F Systems with i.i.d. Components

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Abstract

This paper discusses the reliability of Consecutive-k-out-of-n: F systems with i.i.d. components, and presents efficient algorithms for linear and circular cases. The proposed $O(k^{\log_2 7} \log(n/k))$ algorithms are more efficient and much simpler than the past researches.

1. Introduction

A consecutive-k-out-of-n: F system contains n ordered components which are arranged in a line or a circle. The system fails if and only if at least k consecutive components all fail.

Kontoleon [6] first studied such system, Chiang & Niu [1] used the notation of consecutive-k-out-of-n: F system. Derman, Lieberman, Ross [3] extended the linear system to a circular arrangement. The system can represent telecommunications, microwave stations of a telecom network, oil pipeline, computer ring networks, and other kinds of systems.

The system contains i.i.d. (independent and identically distributed) components which are assumed to be statistically independent, and has the same probability p of working, and $q = 1 - p$ of failing. This paper discusses the reliability of Consecutive-k-out-of-n: F systems with i.i.d. components.

Consider the system with i.i.d. components, Du & Hwang [4] first derived $O(n)$ algorithms to compute the reliabilities of both linear and circular consecutive-k-out-of-n: F systems. And then, Hwang & Wright [5] proposed $O(k^3 \log(n/k))$ algorithms, and Lin [7] derived $O(k^2 \log(n))$ algorithms.

This paper derives $O(k^{\log_2 7} \log(n/k))$ algorithms for computing the reliabilities of both linear and circular consecutive-k-out-of-n: F systems with i.i.d. components. The results are more efficient and much simpler than the past researches.

2. Notation

- n number of components in a system
- k minimum number of consecutive failed components which causes system failure
- p probability that a component works
- q probability that a component fails, $q = 1 - p$
- $R_L(n)$ reliability of linear consecutive-k-out-of-n: F system
- $R_C(n)$ reliability of circular consecutive-k-out-of-n: F system

3. Linear System

Consider linear consecutive-k-out-of-n: F systems with i.i.d. components, the recursive equation $R_L(n)$ for $n > k$, is proposed by Du & Hwang [3]. We can derive the boundary conditions for $n = 0, 1, \dots, k$, as follows.

$$R_L(n) = \begin{cases} R_L(n-1) - pq^k R_L(n-k-1), & n > k \\ 1 - q^k, & n = k \\ 1, & n = 0, \dots, k-1 \end{cases} \quad (1)$$

Let $C = -pq^k$, we can derive the matrix format.

$$\begin{bmatrix} R_L(n) \\ R_L(n-1) \\ \vdots \\ R_L(n-k) \end{bmatrix}_{(k+1) \times 1} = \begin{bmatrix} 1 & \cdots & 0 & C \\ 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix}_{(k+1) \times (k+1)} \begin{bmatrix} R_L(n-1) \\ R_L(n-2) \\ \vdots \\ R_L(n-k-1) \end{bmatrix}_{(k+1) \times 1} \quad (2)$$

$$\text{Let } M = \begin{bmatrix} 1 & \cdots & 0 & C \\ 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix}_{(k+1) \times (k+1)},$$

the follows can be recursively derived.

$$\begin{aligned} & \begin{bmatrix} R_L(n) \\ R_L(n-1) \\ \vdots \\ R_L(n-k) \end{bmatrix}_{(k+1) \times 1} \\ &= M \begin{bmatrix} R_L(n-1) \\ R_L(n-2) \\ \vdots \\ R_L(n-k-1) \end{bmatrix}_{(k+1) \times 1} \\ &= M \left(M \begin{bmatrix} R_L(n-2) \\ R_L(n-3) \\ \vdots \\ R_L(n-k-2) \end{bmatrix}_{(k+1) \times 1} \right) \\ &= MM \begin{bmatrix} R_L(n-2) \\ R_L(n-3) \\ \vdots \\ R_L(n-k-2) \end{bmatrix}_{(k+1) \times 1} \\ &= M^2 \begin{bmatrix} R_L(n-2) \\ R_L(n-3) \\ \vdots \\ R_L(n-k-2) \end{bmatrix}_{(k+1) \times 1} \\ &= \dots = M^{n-k} \begin{bmatrix} R_L(k) \\ R_L(k-1) \\ \vdots \\ R_L(0) \end{bmatrix}_{(k+1) \times 1} \\ &= M^{n-k} \begin{bmatrix} 1-q^k \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{(k+1) \times 1} \end{aligned} \quad (3)$$

$M, M^2, M^3, \dots, M^k$ can be easily derived as follows.

$$\begin{aligned} M &= \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 & C \\ 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 \end{bmatrix}_{(k+1) \times (k+1)}, \\ M^2 &= \begin{bmatrix} 1 & 0 & \cdots & 0 & C & C \\ 1 & 0 & \cdots & 0 & 0 & C \\ 1 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 1 & 0 & 0 \end{bmatrix}_{(k+1) \times (k+1)}, \end{aligned}$$

$$\begin{aligned} M^3 &= \begin{bmatrix} 1 & 0 & \cdots & C & C & C \\ 1 & 0 & \cdots & 0 & C & C \\ 1 & 0 & \cdots & 0 & 0 & C \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}_{(k+1) \times (k+1)}, \dots, \\ M^k &= \begin{bmatrix} 1 & C & \cdots & C & C & C \\ 1 & 0 & \cdots & C & C & C \\ 1 & 0 & \cdots & C & C & C \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & \cdots & 0 & 0 & C \\ 1 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}_{(k+1) \times (k+1)} \end{aligned} \quad (4)$$

For example, $k = 3$, M, M^2 , and M^3 are as follows.

$$\begin{aligned} M &= \begin{bmatrix} 1 & 0 & 0 & C \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \\ M^2 &= \begin{bmatrix} 1 & 0 & C & C \\ 1 & 0 & 0 & C \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \\ \text{and } M^3 &= \begin{bmatrix} 1 & C & C & C \\ 1 & 0 & C & C \\ 1 & 0 & 0 & C \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{aligned} \quad (5)$$

We now evaluate matrix M^{n-k} . Let $n-k = ak + b$, so $a = \lfloor n/k \rfloor, 0 \leq b < k$. M^{n-k} is evaluated as

$$M^{n-k} = M^{ak+b} = [M^k]^a M^b. \quad (6)$$

M^k and M^b can be obtained by Formula (4), computing $[M^k]^a$ requires only $O(\log a) = O(\log(n/k))$ matrix multiplications by Fast Matrix Power Algorithm [7]. Fast Matrix Power Algorithm is presented as follows.

Fast Matrix Power Algorithm

Input: M, n

Output: M^n

Begin

$M_2 \leftarrow I;$

for $r = 1$ to $\log_2 \lceil n \rceil$ do

 If the r th bit of the binary representation of n is 1 then $M_2 \leftarrow M_2 \times M;$

$M \leftarrow M \times M;$

end for;

return $M_2;$

End;

For example, $n = 44$ and $k = 3$, the binary representation of 13 is 1101_2 . M^{44-3} is evaluated as

$$\begin{aligned} & M^{44-3} \\ &= M^{41} \\ &= M^{13 \cdot 3 + 2} \\ &= [M^3]^8 \cdot [M^3]^4 \cdot [M^3] \cdot M^2. \end{aligned} \quad (7)$$

Consider matrix multiplications, Strassen algorithm [2] requires $O(k^{\log_2 7}) \approx O(k^{2.807})$ time for two k -by- k matrices. So the time complexity to compute M^{n-k} is $O(k^{\log_2 7} \log(n/k))$. In Formula (3), $R_L(n)$ can be obtained by dot product of the first row of M^{n-k} and column $[1-q^k, 1, \dots, 1]^T$, it need $O(k)$ time. The total time to compute $R_L(n)$ is $O(k^{\log_2 7} \log(n/k)) + O(k) = O(k^{\log_2 7} \log(n/k))$.

4. Circular System

Consider circular consecutive- k -out-of- n : F systems with i.i.d. components, the recursive equation $R_C(n)$ for $n > k$, is proposed by Du & Hwang [3]. We can derive the boundary conditions for $n = 0, 1, \dots, k$, as follows.

$$R_C(n) = \begin{cases} R_C(n-1) - pq^k R_C(n-k-1), & n > k \\ 1 - q^n, & n = 1, \dots, k \\ k, & n = 0 \end{cases} \quad (8)$$

Let $C = -pq^k$, we can derive the matrix format.

$$\begin{aligned} & \begin{bmatrix} R_C(n) \\ R_C(n-1) \\ \vdots \\ R_C(n-k) \end{bmatrix}_{(k+1) \times 1} \\ &= \begin{bmatrix} 1 & \cdots & 0 & C \\ 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix}_{(k+1) \times (k+1)} \begin{bmatrix} R_C(n-1) \\ R_C(n-2) \\ \vdots \\ R_C(n-k-1) \end{bmatrix}_{(k+1) \times 1} \\ &= M \begin{bmatrix} R_C(n-1) \\ R_C(n-2) \\ \vdots \\ R_C(n-k-1) \end{bmatrix}_{(k+1) \times 1} \\ &= M \left(M \begin{bmatrix} R_C(n-2) \\ R_C(n-3) \\ \vdots \\ R_C(n-k-2) \end{bmatrix}_{(k+1) \times 1} \right) \end{aligned}$$

$$\begin{aligned} &= MM \begin{bmatrix} R_C(n-2) \\ R_C(n-3) \\ \vdots \\ R_C(n-k-2) \end{bmatrix}_{(k+1) \times 1} \\ &= M^2 \begin{bmatrix} R_L(n-2) \\ R_L(n-3) \\ \vdots \\ R_L(n-k-2) \end{bmatrix}_{(k+1) \times 1} \\ &= \dots = M^{n-k} \begin{bmatrix} R_C(k) \\ R_C(k-1) \\ \vdots \\ R_C(0) \end{bmatrix}_{(k+1) \times 1} \\ &= M^{n-k} \begin{bmatrix} 1 - q^k \\ 1 - q^{k-1} \\ \vdots \\ k \end{bmatrix}_{(k+1) \times 1} \end{aligned} \quad (9)$$

Using Fast Matrix Power Algorithm and Strassen algorithm, M^{n-k} can be evaluated in $O(k^{\log_2 7} \log(n/k))$ time. $R_C(n)$ can be obtained by dot product of the first row of M^{n-k} and column $[1-q^k, 1-q^{k-1}, \dots, k]^T$, it need additionally $O(k)$ time. So the total time to compute $R_C(n)$ is also $O(k^{\log_2 7} \log(n/k)) + O(k) = O(k^{\log_2 7} \log(n/k))$.

5. Conclusion

In this paper, we discuss the reliability of Consecutive- k -out-of- n : F systems with i.i.d. components, and propose $O(k^{\log_2 7} \log(n/k))$ algorithms for linear and circular cases. Our results are more efficient and much simpler than the past researches.

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